

Beyond Static Benchmarks: A Post-Leaderboard Evaluation Paradigm for Generative Music and Creative AI

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Music AI Evaluation: Current State

Evaluation Crisis in Generative Music

The current state of evaluation in generative music is methodologically fragmented, relies on ambiguous validation objectives, and builds on systematically misaligned metrics [1]. This has led to a crisis in the field, where the following failures are observed:

- **Validity failure:**
 - Utilized metrics are *unreliable proxies* and fail to measure musical qualities.
- **Leaderboard failure:**
 - Single-score optimization leads to *overfitting* and is *insufficient for assessing a multi-dimensional cultural artifact*.
- **Comparability failure:**
 - *Different metrics, feature representations, and reference data* prevent meaningful comparisons across models.
- **Reproducibility failure:**
 - *Closed-source representations and evaluation scripts* and proprietary data hinder reproducibility.
- **Governance failure:**
 - *Ad-hoc, fragmented evaluation protocols* indicate lack of standardization.

Goals of the Proposed Framework

- **Community-driven ecosystem governance:**
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- **Multi-dimensional evaluation:**
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- **Interpretability:**
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- **Domain adaptability:**
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- **Reproducibility & transparency:**
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Architecture

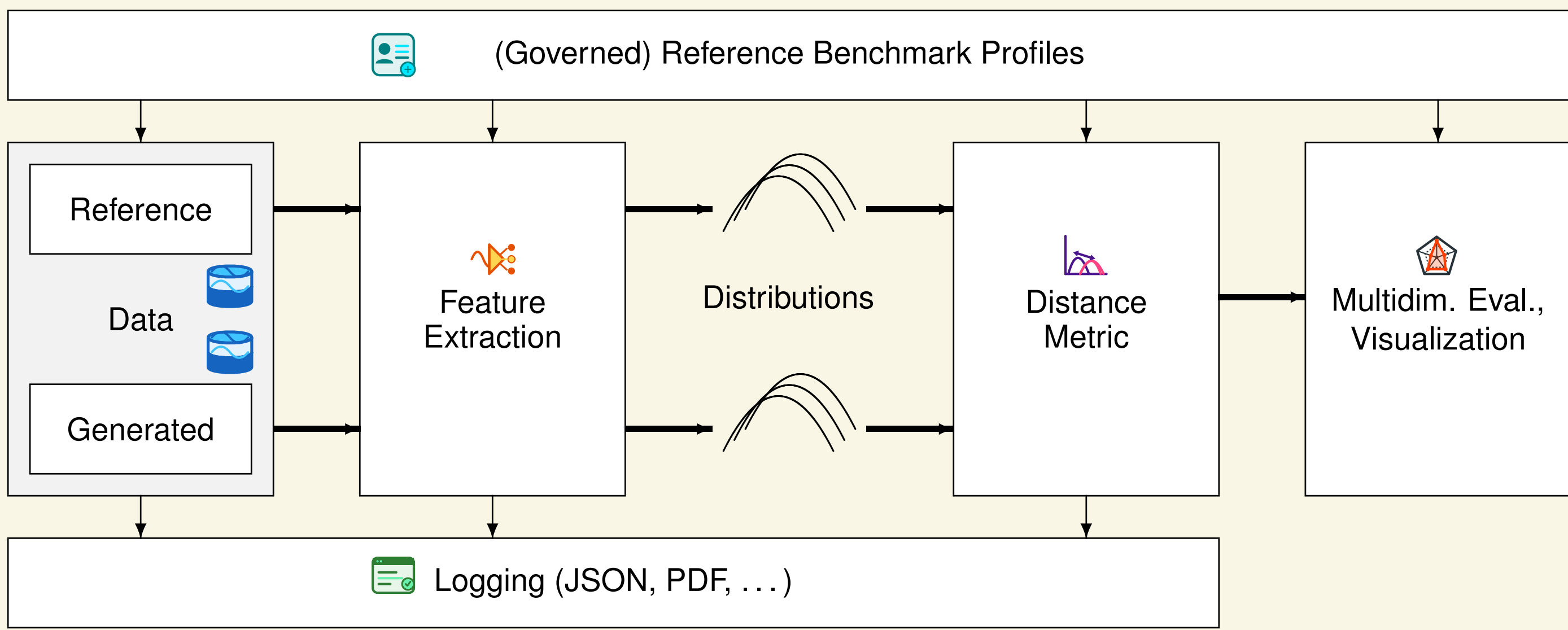


Figure: Proposed architecture for a post-leaderboard evaluation framework for generative music.

- **Reference Benchmark Profiles (RBPs):**
- **Standardized Reference Data:**
- **Standardized Feature Representations:**
- **Standardized Distance Metrics:**
- **Multi-dimensional Results & Visualization:**
- **Logging:**

Governance Structure

Limitations

- Focus on system output:
 - no usability or user experience evaluation
 - no evaluation of the underlying model architecture
 - no distinction between human contributions and fully machine-generated output
- current scope only on objective evaluation of musical output, no subjective evaluation
- no evaluation of
 - explainability
 - bias
 - ethical use of data
 - resource usage

References

Alexander Lerch, Claire Arthur, Nick Bryan-Kinns, Corey Ford, Qianyi Sun, and Ashvala Vinay. Survey on the Evaluation of Generative Models in Music. *ACM Computing Surveys*, 58(4):99:1–99:36, October 2025.